ShramSetu

AI-First Multilingual Welfare Platform for Migrant Workers

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Outline

1. Introduction
2. Problem Statement
3. Literature Review
4. Proposed Solution
5. System Architecture
6. Use Case Diagram
7. Data Flow Diagrams (Level 0 / 0.1)
8. Methodology
9. Functional & Non-Functional Requirements
10. Hardware & Software Requirements
11. Results & Tests
12. Conclusion
13. Future Scope
14. References

INTRODUCTION

Migrant workers fall through the cracks of welfare support

Background

- Welfare information is scattered across official portals and offices.
- Language and digital-literacy barriers limit access.
- Workers rarely know the correct grievance channel.
- General chatbots answer without citing any source.

Motivation: make rights and welfare guidance easy to reach — multilingual, mobile-first, source-grounded.

Scope & Objectives

- One worker-facing access point (PWA, six languages).
- Official-source question answering — every answer cites its document.
- Worker-owned records: document wallet, employment & wage diary.
- Grievance filing routed to jurisdiction-scoped officials.

PROBLEM STATEMENT

The gap the platform closes

Scattered sources

Scheme, law and portal information lives in different departments and formats.

Language barrier

Most official content exists only in English or Hindi.

Untrusted answers

Generic chatbots may give confident but unsupported answers.

Unknown channels

Workers cannot identify where to raise a grievance, or track it afterwards.

ShramSetu: one entry point → answers grounded in indexed official documents → explicit no-match when evidence is insufficient → tracked grievance routing.

LITERATURE REVIEW (1/2)

Prior worker-support systems

Paper	Methodology	Reported evaluation	Gap / relevance
Migrant Care (2025)	Next.js, Convex, Firebase auth, multilingual UI, complaint alerts.	Registration 300 ms; complaint submission 450 ms; language switch < 100 ms.	No voice input or offline mode.
Enhancing Oversight & Support (2026)	Next.js, Python, MySQL; QR profiles, insurance, job search, grievance redressal.	No explicit accuracy metrics; administration-focused.	Limited performance transparency.
Job Bridge (2025)	Web job portal, Flask, MySQL, AI skill matching, Google Translate.	Qualitative improvements reported; no numerical accuracy.	Basic matching; untested scalability.

LITERATURE REVIEW (2/2)

RAG and e-government systems — what they lack

Paper	Methodology	Reported evaluation	Gap / relevance
RAG-Based QA for Personalized Databases (2024)	SQL, Chroma, LangChain, MongoDB history; RAGAS model/embedding comparisons.	Faithfulness 0.51, relevancy 0.86, precision 0.79 (best combo).	Single organisation's data; motivates source-grounded evidence.
E-Government Chatbots (2023)	PRISMA review, deployed chatbot survey, Dialogflow prototypes.	Task completion 100% vs 66.7% portal baseline in one test.	Validation need; motivates tested grounding.
AI-Powered Local Worker Marketplace (2026)	React, Node.js, Supabase, Redis, Socket.IO; rule-based chatbot.	91%+ intent accuracy; response time 340 ms → 82 ms with Redis.	Rule-based only; motivates RAG with citations.

PROPOSED SOLUTION (1/2)

System approach — ask, retrieve, ground, refuse

1. Select language and authenticate (OTP / password, JWT).
2. Ask questions by text or voice.
3. Retrieve relevant passages from indexed official documents (FAISS).
4. Generate an answer constrained to retrieved evidence (similarity gate + BM25/RRF + LLM).
5. Show source metadata — or return an explicit no-match when evidence is insufficient.

Supporting modules

- Encrypted worker document wallet (AES-256-GCM, wrapped keys).
- Portable employment & salary (wage diary) records.
- Grievance submission with validated attachments.
- Category-aware routing to jurisdiction-scoped officials.
- Source traceability + PII-stripped privacy controls.

PROPOSED SOLUTION (2/2)

Core design principles

Mobile-first PWA	React + TypeScript, installable, code-split (entry JS 592 → 406 kB).
Six languages	English, Hindi, Bengali, Telugu, Tamil, Malayalam — parity-tested bundles.
Two separate corpora	Public government knowledge is indexed; private worker documents stay encrypted under the worker's account.
Explicit no-match	Retrieval threshold (0.45) refuses off-topic questions before the LLM; insufficient evidence => 'could not find', never a guess.
Informational support only	Official verification and legal decisions remain with the authorities — stated on every answer.

SYSTEM ARCHITECTURE

Three-service, polyglot architecture (as built)

Frontend · PWA

- React 18 + TypeScript
- Vite + Tailwind
- i18n: 6 languages
- 16 lazy-loaded pages
- Talks only to /api (JWT)

JWT

API server · :8000

- Node 22 + Express 4
- Mongoose 8 → MongoDB
- OTP/JWT auth, rate limits
- Fail-closed jurisdiction scoping
- AES-256-GCM document wallet

X-Internal-Key

AI service · :8100

- Python 3.10 + FastAPI
- FAISS + multilingual embeddings
- 0.45 gate → BM25/RRF → Groq
- STT faster-whisper + TTS local
- PII stripped pre-provider

Design: browser never calls the AI service; provider keys stay server-side; AI service is stateless (no DB).

MongoDB :27017 — workers, grievances, documents, wages, notifications, audit log

External: Groq (LLM) → Gemini fallback · local Whisper STT · local MMS/IndicParler TTS

Requests: Browser —JWT—▶ Express (auth → rate-limit → route) —Mongoose—▶ MongoDB · Express —X-Internal-Key—▶ FastAPI (PII strip → translate → FAISS → gate → Groq)

USE CASE DIAGRAM

Actors and their verified capabilities

Worker

- Register (OTP) · choose language · ask by text or voice · hear answers
- Upload documents (encrypted wallet) · wage diary · QR ID
- File & track grievances · read notifications

Government official

- Dashboard scoped by state / district (fail-closed, 404 on out-of-scope)
- Act on grievances within workflow + SLA
- Send notifications to own jurisdiction only (403 otherwise)

Superadmin

- Full access + audit logs + officials monitoring

RAG assistant (system)

- Retrieve from official corpus · enforce 0.45 gate
- Answer with citations or refuse (grounded = false)

Every edge maps to a tested route in server/src/routes/ (49 server tests · 92 AI tests · 50 frontend tests).

DATA FLOW — LEVEL 0

Context diagram



DATA FLOW — LEVEL 0.1

Main processes inside the platform

1 · Access & identity	OTP / JWT authentication, language profile
2 · Assistant (RAG)	PII strip → translate → FAISS → 0.45 gate → Groq answer + citations
3 · Records	Document wallet (AES-256-GCM) · wage diary · QR ID
4 · Grievances	Category capture → jurisdiction-scoped official queue → status + SLA
5 · Notifications	In-app inbox scoped per worker · audit log for officials

METHODOLOGY

Ten steps — as implemented

1. **Language + access**
   choose language; OTP / JWT authentication
2. **Query intake**
   typed question or faster-whisper voice transcription
3. **Knowledge prep**
   extract, clean, chunk, index with provenance (source URL, verified date, status)
4. **Semantic retrieval**
   embed query, FAISS top-k over 62 active docs
5. **Evidence check**
   0.45 similarity gate; explicit no-match when below
6. **Grounded response**
   LLM answers from context with source, department, freshness
7. **Worker records**
   encrypted wallet + employment + wage entries
8. **Grievance workflow**
   category, description, evidence → jurisdiction-scoped official queue
9. **Persistence & security**
   authenticated APIs; owner-based access; PII stripped pre-provider
10. **Evaluation**
    retrieval, grounding, refusals, multilingual, security, latency

REQUIREMENTS

Functional & non-functional

Functional

- Register / authenticate workers and officials; preferred language.
- Answer text and voice questions from indexed official sources.
- Show source details; refuse explicitly on insufficient evidence.
- Secure upload and management of personal documents.
- Maintain employment and salary (wage) records.
- Evidence-backed grievances routed by category + jurisdiction.

Non-functional

- Security — JWT sessions, hashed credentials, owner-based access, fail-closed scoping.
- Privacy — private files separate from the public RAG corpus; PII stripped before providers.
- Reliability — refuse != failure; health liveness/readiness split.
- Usability — mobile-first, multilingual, voice accessible.
- Maintainability — modular client / API / data / AI services.
- Traceability — source metadata and index dates preserved.

HARDWARE REQUIREMENTS

Enough to build and demo on commodity hardware

- Client device — Android / iOS phone or any modern desktop browser (PWA).
- Development system — 64-bit multi-core PC (React/TS + Python builds + model loading).
- Backend host — CPU-only server: Node 22 (API) + Python 3.10 (AI service) + MongoDB.
- Storage — persistent volume for app data, uploaded documents (AES-256-GCM at rest) and the indexed corpus / FAISS index.
- Network — internet for provider calls (Groq) ; STT and TTS run locally, no key needed.

SOFTWARE REQUIREMENTS

Stack as built (each line verified in the repo)

Web client	React 18 + TypeScript + Vite + Tailwind CSS — PWA, code-split
Backend / API	Node.js 22 + Express 4 + JWT auth + helmet + per-route rate limits
Persistence	MongoDB (Mongoose 8) — embedded dev instance or Docker volume in production
Retrieval & AI	LangChain, Sentence-Transformers (multilingual MiniLM), FAISS, Groq LLM (Gemini fallback)
Speech	faster-whisper STT (local, per-language checkpoints) · AI4Bharat IndicParler-TTS with local MMS VITS fallback
Notifications	In-app worker inbox + government broadcast (FCM token endpoint ready for real push)
Quality tooling	pytest (92), vitest (49 server + 50 frontend), 34 live E2E checks, RAG eval harness

RESULTS

Retrieval evaluation — headline numbers

hit@1

94 %

46 / 49 factual questions answered from the right document

hit@3

98 %

48 / 49 — correct document inside the top three

Refusals

12 / 13

off-topic & prompt-injection questions refused at the 0.45 gate (92 %)

Corpus & harness

62 active official documents indexed (11 repealed, marked superseded, kept out of retrieval) · multilingual 70-question suite across all six languages · hybrid dense + BM25/RRF ranking · reproducible run: python eval/run_eval.py (~30 s, no API key needed in retrieval-only mode)

RESULTS

Safety, tests and honest limits

End-to-end safety (live LLM)

10 / 10: every off-topic, prompt-injection and compound 'reveal your secrets + do X' request refused by the grounded flag with zero system errors (RESULTS_FULL.md).

Test suites (all green in CI)

Server 49 · AI service 92 · frontend 50 · 34 live end-to-end checks incl. wage round trip + paid-day lock + jurisdiction 403/404. CI gates all four.

Honest limits — stated before you ask

- The one remaining hit@3 miss is a measured translation failure (MGNREGA name mangled) — hand-translated query retrieves the right doc at 0.641 top-1; fixed via translation, not by lowering the 0.45 gate.
- 8 in-domain 'unanswerable' questions are refused by the LLM's grounded flag; retrieval-only mode cannot measure them (live-LLM run pending in chunks).
- No real SMS gateway (dev OTP prints to terminal), in-memory rate limits, ~30 s cold start bounded at 90 s.

CONCLUSION

One platform, evidence-first

- Every AI answer is either grounded in a cited, dated official document or explicitly refused — no guesses.
- Worker records live encrypted under the worker's own account (AES-256-GCM, wrapped keys).
- Grievances reach officials with enforceable, tested jurisdiction over the filing location.
- Six languages, voice in/out, PII stripped before any external provider call.
- The platform facilitates access — it deliberately does not replace official verification, legal advice, or authority decisions.

"It makes the evidence reachable — not the decision for you."

FUTURE SCOPE

What we would do next, in order

Translation layer for scheme names	entity-aware glossary (MGNREGA, PMSBY…) across six languages — turns the one known hit@3 miss into a hit.
Live-LLM eval completion	finish --full run so the 8 in-domain refusals have a measured number.
Reranking + stronger multilingual embeddings	cross-encoder reranking of the FAISS top-24 pool.
Automated corpus refresh	scheduled re-ingest with freshness alerts + superseded-law drift checks.
Real integrations	SMS gateway for OTPs, FCM push, verified government API where available, offline sync.
Browser E2E suite + field evaluation	Playwright over the OTP → grievance → resolution loop; grounding/usability study with workers.

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